Functional Annotation Report: OpgH (Q88D04, PP_5025) in *Pseudomonas putida* KT2440

Gene: *opgH* (synonym *mdoH*; ordered locus PP_5025) **Protein:** Glucans biosynthesis glucosyltransferase H **UniProt:** Q88D04 **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440) **EC:** 2.4.1.– | **Family:** Glycosyltransferase family 2 (GT2), OpgH subfamily | **Pfam:** PF00535 (Glycos_transf_2) | **HAMAP:** MF_01072

Summary

OpgH (Q88D04, encoded by PP_5025) in *Pseudomonas putida* KT2440 is a **polytopic inner (cytoplasmic) membrane glycosyltransferase of family GT2 (EC 2.4.1.–)** that catalyzes the committed, backbone-polymerizing step of **osmoregulated periplasmic glucan (OPG)** biosynthesis. Using cytoplasmic **UDP-glucose** as the sugar donor, OpgH assembles a β-linked polyglucose chain at the cytoplasmic face of the inner membrane and is topologically positioned to translocate the nascent glucan into the periplasm. There, its operon partner **OpgG** — recently defined as a β-1,2-glucanase of the new glycoside hydrolase family GH186 — processes and shapes the polymer. In *Pseudomonas*, the mature OPGs are neutral, highly branched glucans of 6–13 glucose units linked by **β-1,2 bonds (backbone) and β-1,6 bonds (branches)**, whose abundance rises at low osmolarity — the defining hallmark of OPGs.

The physiological role of the OpgH product is that of a **periplasmic envelope component involved in osmotic and periplasmic-volume adaptation, membrane and detergent/antibiotic resistance, and Rcs-mediated envelope-stress signaling**. Loss of OPG synthesis (via *opgH*/*mdoH* mutation) produces a pleiotropic envelope phenotype, sensitizes cells to SDS, perturbs the periplasmic Donnan potential, and de-represses colanic-acid capsule synthesis through the RcsC sensor. In enterobacteria and *Pseudomonas syringae*, the *opgGH* (*mdoGH* / *hrpM*) locus is additionally required for full virulence, linking periplasmic glucans to host interaction. Beyond its enzymatic role, OpgH **moonlights** in *E. coli* as a nutrient-dependent antagonist of the cell-division protein **FtsZ**, using UDP-glucose as a proxy for nutritional status to couple cell size to central metabolism.

An important disambiguation was resolved during this investigation: OpgH is **distinct** from a separately encoded *P. putida* KT2440 BcsA-like "orphan" protein that synthesizes **cyclic-β-1,3-glucan** via a fused GT2–GH17 architecture. OpgH is a single-GT2-domain OpgH-subfamily enzyme producing β-1,2/β-1,6 OPGs, whereas the orphan synthase is a two-domain enzyme producing a different polymer. The two therefore represent **two independent periplasmic glucan pathways** in the same organism. Finally, note that the specific functional assignment for PP_5025 rests on **strong orthology** to biochemically characterized *E. coli* and *Pseudomonas* OpgH/MdoH proteins rather than on direct experimental study of the *P. putida* protein itself — the annotation is confident but inferential, and gene symbol, organism, GT2/OpgH family, and Pfam PF00535 all match the OPG glucosyltransferase, confirming this is the correct target.

Key Findings

F001 — OpgH is a GT2-family inner-membrane glucosyltransferase that polymerizes the OPG glucose backbone from UDP-glucose

P. putida OpgH belongs to **glycosyltransferase family 2** (Pfam PF00535; HAMAP rule MF_01072), the family that defines its enzymatic class. The reaction it catalyzes is the transfer of glucose from the nucleotide-sugar donor **UDP-glucose** onto a growing glucan chain, building the polysaccharide backbone of osmoregulated periplasmic glucans. This activity is best defined through its biochemically characterized orthologs. In *Yersinia pseudotuberculosis*, the *opgGH* operon "encodes glucosyltransferases that synthesize osmoregulated periplasmic glucans (OPGs) from UDP-glucose, using acyl carrier protein (ACP) as a cofactor" PMID: 26150539 — this single statement pins down substrate (UDP-glucose), product (OPGs), cofactor (ACP), and enzymatic class. Direct biochemical evidence comes from *E. coli*, where "mutants lacking MdoH are deficient in a glucosyltransferase activity assayed in vitro" PMID: 9352918, and where MdoH "presence was shown to be necessary for normal glucosyl transferase activity" PMID: 7934824. Because Q88D04 is a clear ortholog carrying the same catalytic GT2 domain and HAMAP signature, the same reaction is assigned to the *P. putida* enzyme.

Reaction (backbone synthesis): UDP-glucose + $[\beta\text{-glucan}]_n \rightarrow$ UDP + $[\beta\text{-glucan}]_{n+1}$. **Substrate specificity:** sugar donor = UDP-glucose; acceptor = the elongating β-glucan chain; the product backbone is β-1,2-linked glucose (see F003, F004). **Enzyme classification:** GT2 family (nucleotide-diphospho-sugar transferase fold, IPR029044); EC 2.4.1.– hexosyltransferase.

F002 — OpgH is a polytopic inner-membrane protein with a large cytoplasmic catalytic domain positioned to translocate nascent glucan to the periplasm

The membrane topology of the OpgH ortholog MdoH (847 aa in *E. coli*) was experimentally mapped using **51 different MdoH–β-lactamase fusions**, which showed that "the MdoH protein crosses the cytoplasmic membrane eight times, with the N and C termini in the cytoplasm" PMID: 9352918. Critically, a large **~310-amino-acid cytoplasmic domain** lies between the second and third transmembrane segments and carries the catalytic (glucosyltransferase) function — consistent with the enzyme drawing on cytoplasmic UDP-glucose. The topological model further "suggests that this protein could be directly involved in the **translocation of nascent polyglucose chains to the periplasmic space**" PMID: 9352918, implying a dual polymerization/translocation role. In *Pseudomonas*, the ortholog is likewise "a 97 kDa protein spanning the cytoplasmic membrane" PMID: 7934824. This places OpgH's catalysis on the cytoplasmic face of the inner membrane, with the polymer destined for the periplasm. The partner OpgG is a periplasmic ~56 kDa protein that matures the glucan there (F003).

F003 — OpgH acts with the periplasmic partner OpgG; OpgG/OpgD are β-1,2-glucanases (new family GH186) that shape the glucan

opgH is the second gene of the **opgGH (mdoGH) operon**, and OpgG is a ~56 kDa **periplasmic** protein whose function was long undefined PMID: 7934824. Recent structural and functional work resolved it: analyses of *E. coli* OpgG and OpgD "revealed that these proteins are **β-1,2-glucanases** with remarkably different activity from each other, establishing a new glycoside hydrolase family, **GH186**" PMID: 37735577. Because OpgG hydrolyzes β-1,2 linkages, the backbone made by OpgH is inferred to be **β-1,2-linked glucose**, which OpgG then trims/shapes in the periplasm. The synthesis–hydrolysis logic is reinforced in *Caulobacter*, where OpgH (synthesis) and BglX (hydrolysis) together define an OPG pathway producing a **cyclic hexamer of glucose** PMID: 36516849. Thus OpgH does not act alone: it initiates and polymerizes, while periplasmic glucanases (OpgG/OpgD, or BglX) determine final chain length and architecture. Downstream substituent transferases (e.g., OpgC succinylation in enterobacteria; PMID: 26790533) may decorate the backbone in some species.

F004 — In *Pseudomonas*, OPGs are branched β-1,2/β-1,6 glucans, and the *opgGH*/*mdoGH* locus corresponds to the *hrpM* pathogenicity locus

Chemical characterization of the periplasmic glucans of *Pseudomonas syringae* pv. *syringae* showed they are "**neutral, unsubstituted, and composed solely of glucose**," 6–13 glucose units in size, and that "the glucans are **linked by beta-1,2 and beta-1,6 glycosidic bonds**" — a highly branched structure with a β-1,2 backbone and β-1,6 branches PMID: 7961404. These glucans are osmoregulated: "these periplasmic glucans were more abundant when the osmolarity of the growth medium was lower" PMID: 7961404. This defines the exact product class expected for *P. putida* OpgH — a neutral, branched pseudomonad-type glucan, contrasting with the anionic, substituted *E. coli* MDO. Furthermore, the *E. coli mdoGH* operon shares "**a considerable homology (69% identical nucleotides out of 2816)**" with the two genes at the *hrpM* locus of *Pseudomonas syringae* pv. *syringae*, a locus required for pathogenicity and the plant hypersensitive reaction PMID: 7934824. This directly links the glucosyltransferase operon to a pseudomonad host-interaction function.

F005 — OPGs made by OpgH are periplasmic envelope components required for osmotic adaptation, membrane/detergent resistance, and Rcs-mediated signaling

OPGs are periplasmic oligosaccharides that are "modulated during osmotic fluxes" and constitute intrinsic envelope components (review PMID: 28593831). Their loss is pleiotropic: "mutants defective in OPG synthesis have a highly pleiotropic phenotype, indicative of an overall alteration of their envelope properties" PMID: 10779706. A precise phenotype is detergent resistance: "wild-type *E. coli* MC4100 grew in the presence of **10% SDS** whereas isogenic *mdoA* and *mdoB* mutants could not grow above **0.5% SDS**" PMID: 12798996 — a 20-fold sensitization. OPGs also shape periplasmic physical chemistry: as highly anionic (in enterobacteria) polymers they generate a **Donnan potential** that concentrates cations and contributes to periplasmic volume regulation PMID: 12379178, and MDOs are proposed to "play a key role in periplasmic volume regulation at low-to-moderate osmolality" PMID: 10733957. Finally, periplasmic OPG levels feed into envelope-stress signaling: inactivation of *mdoH* increases colanic-acid capsule via the Rcs system because "the periplasmic levels of MDOs act to signal RcsC to activate *cps* expression" PMID: 9352941. Reviews connect periplasmic OPGs to signal transduction that indirectly regulates virulence, and secreted OPGs to antibiotic and host-cell interactions PMID: 26265506.

F006 — OpgH moonlights as a nutrient-dependent FtsZ antagonist that couples cell size to central metabolism

Beyond making glucan, OpgH has a second, "moonlighting" function established in *E. coli*: it is "a **nutrient-dependent regulator of *E. coli* cell size**." Under nutrient-rich growth, "OpgH localizes to the nascent septal site, where it **antagonizes assembly of the tubulin-like cell division protein FtsZ**, delaying division and increasing cell size," a mechanism consistent with OpgH sequestering FtsZ from growing polymers PMID: 23935518. The molecular link to metabolism is UDP-glucose: the work highlights "the conservation of uridine diphosphate glucose as a proxy for nutrient status and the use of **moonlighting enzymes to couple growth rate-dependent phenomena to central metabolism**" PMID: 23935518. This is independently corroborated: "OpgH also **sequesters FtsZ** in order to regulate cell size according to nutrient availability," and Δ*opgGH* cells are smaller PMID: 26150539. Whether the *P. putida* ortholog retains this moonlighting role has not been directly tested, but the shared UDP-glucose-binding GT2 architecture makes it plausible.

F007 — OpgH is distinct from the co-existing *P. putida* KT2440 cyclic-β-1,3-glucan synthase (a BcsA-like "orphan"); two separate periplasmic glucan pathways

This finding resolves a potential misidentification. *P. putida* KT2440 encodes a **BcsA-like "orphan" protein** that bioinformatic and structural modeling describes as "a two-domain transmembrane ovoid-like structure ... with a periplasmic **glycosyl hydrolase family GH17 domain** linked via a transmembrane region to a **cytoplasmic glycosyltransferase family GT2 domain**" PMID: 37267309. The proposed mechanism is that "the **GT2 domain synthesises β-(1,3)-glucan** that is transferred to the GH17 domain where it is cleaved and cyclised to produce **cyclic-β-(1,3)-glucan (CβG)**" PMID: 37267309. This enzyme shares <41% identity with cellulose synthase BcsA and is **architecturally and product-wise distinct from OpgH**: OpgH is a single-GT2-domain, OpgH-subfamily membrane glucosyltransferase (HAMAP MF_01072; PF00535) that produces β-1,2/β-1,6 osmoregulated periplasmic glucans in partnership with periplasmic OpgG. Thus, *P. putida* KT2440 possesses **two separate periplasmic glucan pathways**: the OpgGH OPG pathway (branched β-1,2/β-1,6 glucans) and the orphan CβG pathway (cyclic β-1,3-glucan). Q88D04/PP_5025 is the OPG pathway enzyme, and the two should not be conflated.

Mechanistic Model / Interpretation

The evidence assembles into a coherent, spatially organized model of OpgH function at the *P. putida* inner membrane:

```
 CYTOPLASM                            INNER MEMBRANE                    PERIPLASM
--------------------------------------------------------------------------------------

 UDP-glucose ---->  [ OpgH cytoplasmic GT2 domain ]
  (donor, and       (~310 aa catalytic loop between TM2 and TM3)
  nutrient proxy)          |
                           | polymerize beta-1,2 backbone
                           v
                   ====== OpgH (8 TM segments) ======  --translocate-->  nascent
                           N- and C-termini cytoplasmic                    beta-glucan
                                                                              |
                                                                              v
                                                              [ OpgG / OpgD : GH186 beta-1,2-gluca
                                                               trims, shapes, branches (beta-1,6)
                                                                              |
                                                                              v
                                                              Mature OPG: neutral, branched
                                                              beta-1,2 / beta-1,6 glucan (6-13 Gl
                                                              Osmoregulated (up at low osmolarity)

 MOONLIGHTING (nutrient-rich):
 UDP-glucose high --> OpgH at septum --> sequesters FtsZ --> delays division --> larger cells
```

Primary function (enzymatic). OpgH is the backbone-building enzyme of OPG biosynthesis. It binds cytoplasmic UDP-glucose at its large cytoplasmic GT2 domain (F001, F002) and transfers glucose to a growing chain, forming a β-1,2 backbone (F003, F004). Its eight-times-membrane-spanning topology positions it to hand the nascent chain across the inner membrane into the periplasm (F002).

Maturation (partner enzymes). OpgG (and its paralog OpgD), acting as a periplasmic GH186 β-1,2-glucanase, processes the nascent polymer to yield the final OPG species (F003). In *Pseudomonas*, the finished product is a neutral, branched β-1,2/β-1,6 glucan of 6–13 glucose units, more abundant at low osmolarity (F004).

Localization. Catalysis begins on the **cytoplasmic face of the inner membrane**; the finished polymer resides in the **periplasm** where it performs its physiological work (F002, F003, F005).

Downstream physiology. As a bulky, abundant periplasmic solute, OPGs regulate periplasmic osmotic/volume balance and Donnan potential, maintain envelope integrity (SDS/antibiotic resistance), and, through their concentration, act as an input to Rcs envelope-stress signaling that governs capsule, motility, biofilm, and — in pathogens — virulence (F005). In *Pseudomonas syringae*, the operon corresponds to the *hrpM* pathogenicity locus (F004).

Metabolic coupling (moonlighting). Because OpgH senses UDP-glucose, it doubles as a nutrient gauge; under rich conditions it sequesters FtsZ at the septum to enlarge cells, coupling division timing to central metabolism (F006).

The following table summarizes the two distinct glucan systems in *P. putida* KT2440 and situates OpgH:

Feature	OpgH / OpgGH (PP_5025, Q88D04)	Orphan BcsA-like synthase
Architecture	Single GT2 domain, 8 TM segments	Fused GT2 (cytoplasmic) + GH17 (periplasmic)
Product	Branched β-1,2 / β-1,6 OPGs (6–13 Glc)	Cyclic β-1,3-glucan (CβG)
Donor	UDP-glucose	UDP-glucose
Partner	Periplasmic OpgG/OpgD (GH186 glucanase)	Intramolecular GH17 domain
Regulation	Osmoregulated (↑ low osmolarity)	Not characterized
Evidence	Orthology + biochemistry of MdoH/OpgH	Bioinformatics/structural modeling

Evidence Base

PMID	Title (abbrev.)	How it supports the annotation
9352918	Topological analysis of MdoH	Direct biochemistry: MdoH mutants lack glucosyltransferase activity; 8-TM topology; large cytoplasmic catalytic domain; implicated in glucan translocation
26150539	*opgGH* in *Yersinia*	States substrate (UDP-glucose), product (OPGs), cofactor (ACP); confirms FtsZ-sequestration/cell-size role
7934824	*mdoA*/*hrpM* homology	MdoH necessary for glucosyltransferase activity; 97 kDa membrane protein; OpgG periplasmic; links operon to *Pseudomonas hrpM* pathogenicity
37735577	OpgG/OpgD enzymology	Defines OpgG/OpgD as β-1,2-glucanases (new family GH186), implying β-1,2 backbone
7961404	*P. syringae* periplasmic glucans	Chemical structure: neutral, β-1,2/β-1,6-linked, 6–13 Glc, osmoregulated
36516849	EstG/OpgH/BglX in *Caulobacter*	OpgH (synthesis) + glucanase (hydrolysis) pathway; cyclic hexamer OPG product
12798996	MDOs and SDS resistance	*mdoH* mutants: 20-fold SDS sensitization — envelope-integrity role
9352941	*mdoH* and colanic acid	Periplasmic MDO levels signal via RcsC to control capsule
23935518	Moonlighting enzyme/cell size	OpgH sequesters FtsZ; UDP-glucose as nutrient proxy
12379178	Periplasmic Ca^{2+}/ Donnan	MDOs generate Donnan potential concentrating periplasmic cations
10733957	Turgor/periplasmic water	MDOs proposed key to periplasmic volume regulation
10779706	OPGs in Proteobacteria	OPG-deficient mutants have pleiotropic envelope alteration
28593831	OPG review	OPGs modulated during osmotic fluxes; core envelope osmoadaptation
26265506	OPG biological roles review	Periplasmic OPGs in signal transduction/virulence; secreted OPGs in antibiotic/host interaction

PMID	Title (abbrev.)	How it supports the annotation
37267309	BcsA-like orphan synthases	Distinguishes *P. putida* cyclic-β-1,3-glucan synthase from OpgH
26790533	*opgC* succinylation	Backbone (not substituents) required for virulence in *Dickeya*; contextualizes OPG modification
7814331	UDP-glucose as signal	UDP-glucose as intracellular signal (σS control), context for OpgH's nutrient-sensing

Consistency of the evidence. The enzymatic assignment (GT2 glucosyltransferase using UDP-glucose) is supported by both sequence/family criteria and direct in vitro biochemistry of orthologs (3 independent lines: 9352918, 26150539, 7934824). The product chemistry for *Pseudomonas* specifically is established (7961404). Localization is supported by experimental topology mapping (9352918). The downstream physiology is multiply corroborated across osmoadaptation, envelope integrity, Donnan/volume, and Rcs signaling. No retrieved paper contradicts the core annotation.

Supported and Refuted Hypotheses

Supported - OpgH is a GT2 inner-membrane glucosyltransferase using UDP-glucose to build the OPG β-glucan backbone (26150539, 9352918, 7934824). - OpgH is a polytopic inner-membrane protein (8 TM) with a cytoplasmic catalytic domain; its product is periplasmic (9352918). - OpgH acts in an operon/pathway with the periplasmic β-1,2-glucanase OpgG (37735577, 7934824). - Pseudomonad OPGs are neutral branched β-1,2/β-1,6 glucans, osmoregulated (7961404). - OpgH product underlies envelope integrity, osmoadaptation, Rcs signaling, and pathogenicity (12798996, 9352941, 10779706, 7934824). - OpgH moonlights as an FtsZ-sequestering cell-size regulator (23935518).

Refuted / Excluded - OpgH is **not** the *P. putida* cyclic-β-1,3-glucan "orphan" BcsA-like synthase (a separate GT2/GH17 enzyme; 37267309). - OpgH is **not** the periplasmic glucanase — that role belongs to OpgG/OpgD (GH186; 37735577).

Limitations and Knowledge Gaps

1. **No direct study of PP_5025.** Every mechanistic claim about Q88D04 is transferred by orthology from *E. coli* MdoH/OpgH, *Yersinia*, *Caulobacter*, and *P. syringae*. No paper reports purification, kinetics, gene knockout, or structural determination of the *P. putida* KT2440 OpgH itself. The assignment is confident but inferential.
2. **Cofactor and precise mechanism.** The ACP-cofactor requirement is cited from the *Yersinia opgGH* description (26150539); the exact role of ACP, the initiation/priming mechanism, and processivity of the *P. putida* enzyme are not directly established.
3. **Product structure in *P. putida* specifically.** The β-1,2/β-1,6 branched structure is documented for *P. syringae* (7961404); *P. putida* KT2440 OPGs have not been chemically characterized here. Whether substituents (e.g., succinyl, as in *Dickeya* via OpgC, 26790533) decorate *P. putida* OPGs is unknown — although pseudomonad OPGs are reported neutral/unsubstituted.
4. **Moonlighting role unverified in *P. putida*.** FtsZ antagonism/cell-size control is demonstrated in *E. coli* (23935518) and *Yersinia* (26150539); its conservation in *P. putida* is plausible but untested.
5. **Signaling wiring may differ.** Rcs-mediated capsule control (9352941) and Donnan-potential effects (12379178) derive from enterobacteria whose OPGs are anionic; *Pseudomonas* OPGs are neutral/unsubstituted (7961404), so downstream physiology may be quantitatively different.
6. **Two-pathway coexistence.** The relationship, regulation, and possible functional redundancy between the OpgGH OPG pathway and the orphan cyclic-β-1,3-glucan pathway (37267309) in *P. putida* is unexplored.

Proposed Follow-up Experiments / Actions

1. **Targeted knockout of PP_5025 (*opgH*) in *P. putida* KT2440** with phenotyping for: low-osmolarity growth, SDS/antibiotic sensitivity, and cell-size distribution — directly testing F005/F006 in the native organism.

2. **Chemical characterization of *P. putida* KT2440 OPGs** (isolation + NMR/mass spectrometry of linkage composition and degree of polymerization) to confirm the β-1,2/β-1,6 branched structure and check for substituents.
3. **In vitro reconstitution:** express and purify the OpgH cytoplasmic GT2 domain (or full-length in nanodiscs) and assay UDP-glucose-dependent glucosyltransferase activity, including the ACP-cofactor dependence.
4. **Reconstitute the OpgH–OpgG pair** to demonstrate coupled synthesis–maturation and determine final chain-length control by the GH186 partner.
5. **Test FtsZ-sequestration/cell-size moonlighting in *P. putida*** via localization (fluorescent OpgH), FtsZ co-localization, and cell-size measurement across nutrient conditions.
6. **Genetically dissect pathway crosstalk** between OpgGH and the orphan cyclic-β-1,3-glucan synthase (single and double mutants) to define whether the two periplasmic glucan systems are redundant, complementary, or independently regulated.
7. **Structural determination** (cryo-EM) of *P. putida* OpgH to confirm the 8-TM topology and visualize the UDP-glucose-binding catalytic domain and putative translocation path.

Report generated from a 3-iteration autonomous investigation comprising 7 confirmed findings and 20 reviewed papers. The functional annotation of OpgH (Q88D04 / PP_5025) rests on strong orthology to biochemically characterized MdoH/OpgH proteins; direct experimental study of the P. putida protein is the principal outstanding gap.
